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Understanding the development patterns of informal settlements in Nairobi

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Abstract

Informal settlements are a widespread phenomenon in African cities. This study explores the development patterns of informal settlements in Nairobi, Kenya, with a focus on the physical patterns, affordability of plots and rented rooms, and characteristics of developers. The results of the study were as follows:

1. **Spatial characteristics and conditions in informal settlements are not uniform. We identified squatter areas with iron-sheet houses and narrow streets, planned areas with high-rise buildings made of concrete blocks arranged in a perfect array, and intermediate conditions.**
2. **Housing plot prices are unaffordable for most residents, while room rentals are generally fixed in accordance with the area and offer low-income residents a wide range of choices.**
3. **The owners of some structures reside in their own structure, and others are non-residents who build and rent structures for profit. Therefore, informal settlements are not so much a place to live, but rather an attractive investment opportunity for structure owners or developers.**

Keywords
Informal settlement, development pattern, Nairobi, Africa

1. Introduction

Africa is one of the most rapidly urbanizing area in the world. In 1960, only 15% of the total population in the continent was living in urban area, the number increased up to 40% in 2010, and is predicted to be 60% in 2050.[1] Even with this rapid urbanization, urban planning works in a small part of the city, such as central business district and high-income residential area. Therefore, it is estimated that 30%-70% of the African urban population live in informal settlements, which is considered to be a response to the failure of government controlled urban planning systems to meet the housing needs of urban populations. According to Berner, up to 85% of new housing stock is produced in an extra-legal manner.[2] Not only the so-called "slums," characterized by fragile and densely built housing, lack of physical infrastructure and social services, and poor inhabitants but also "decent" residential areas, with seemingly formal planning and residential standards and middle- to high-income inhabitants, have also been produced in an extra-legal manner.[3] Informal settlements are a widespread phenomenon across socioeconomic statuses in urban Africa.

The emergence of informal settlements in African cities can be traced back to the colonial era, when colonial urban planning divided the urban space into two distinct zones: a "European" space that enjoyed a high level of urban infrastructure and services and an "indigenous" space that had marginal services.[4] Indigenous space, or informal settlements, became a prevailing mode of urban development across Africa in the wave of rapid urbanization after World War II. In the 1960s, as urbanization was addressed in the theory of modernization, informal settlements were seen as an obstacle to building modern cities. The dominant idea was that informal settlements should be eradicated from urban areas by government-led capital-intensive housing provisions, which ended in failure. In the early 1970s, a new paradigm and urban policy were provided by John Turner, a British architect who argued that housing was best provided and managed by those who were to dwell in it rather than being centrally administered by the government.[5] He showed that self-designed settlements by local groups worked better as people were experts on their own situations and should be given the "freedom to build." Within this framework, the government and the private sector act as enablers.

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De Soto proposed the importance of land tenure formalization in his book, "The Mystery of Capital" published in 2000.[6] According to de Soto, the main cause of poverty in developing countries is the continuing lack of access to formal property rights among low-income people. An important idea concerning land tenure formalization is that granting property titles are often presented as a means to provide credit-constrained households with better access to mortgaged credit, thus stimulating investment in business activities. In this view, if the low incomes are to gain access to the benefits of capitalism, their assets must be registered and integrated into unified property systems. The effect of land tenure formalization has been verified in a range of fields of study worldwide. Durand-Lasserve and Selod,[7] and Payne et al.[8] reviewed the literature widely whether land titling programs have achieved the benefits claimed by de Soto and found that they have generally failed to do so.[Note 1] The negative aspects of land tenure formalization,[9] and social and economic difficulties in implementing formalization[10,11] have been also identified as critical challenges.

Previous studies have focused on how to incorporate informal settlements into formal urban planning framework, premised on a legal dichotomy between formal and informal. On the other hand, some recent researches have attempted to reconceptualize informal settlements by releasing from such a dichotomy and clarifying aspects that were not visible in the legal framework. It can be said that it is an attempt to understand the urbanization phenomenon in developing cities including African cities more fundamentally. Durand-Lasserve and Selod showed that the exact nature and content of land tenure is not determined in the paradigm of law but should primarily be viewed as a social relationship involving a complex set of rules that governs land use and ownership.[7] Also, Nkurunziza argued that the processes through which households in these settlements access housing land are not anarchic but are structured and regulated by some form of social ordering.[12] Their success in delivering large quantities of housing land is attributed to the social legitimacy they command, evidenced by the general acceptance and respect they enjoy from those whose relationships they regulate.

The purpose of this study is to empirically clarify the real situation of the informal urbanization process in African cities. There is a growing consensus in the field of research that informal settlements play an important role in providing and managing housing to meet the needs of urban populations, and are not as chaotic as they are often portrayed but are structured and regulated by some form of social ordering. Unless ongoing land development activities or the nature of the unique local system are closely monitored and understood, it will be socially and economically costly to retrofit. Understanding the nature of self-designing activities in informal settlements could offer a fresh perspective to housing policy addressing informal settlements. At the onset, the objective of this research is to understand the development patterns of informal settlements in Nairobi, Kenya, focusing on the spatial characteristics and conditions, the affordability of housing plots and rooms, and the characteristics of developers in informal settlements.

2. Growth of Informal Settlements

Nairobi originated in the late 1890s as a colonial railway settlement. Built for white settlers, the colonial authorities restricted Africans from moving into Nairobi, and Africans without a "pass" or permission to be in town were forcefully removed to maintain the desired order. Colonial authorities feared that African settlements, or informal settlements, would

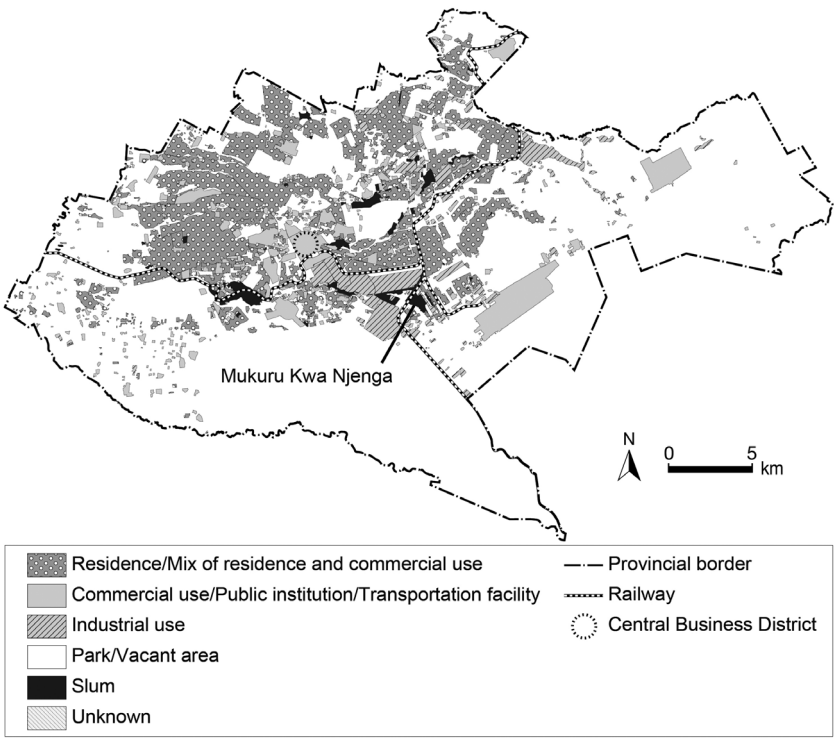

Figure 1. Nairobi land use Source: Kajihara, 2013.

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be hotbeds of crime and disease. However, irrespective of the restriction of African urbanization, many Africans from rural areas moved into Nairobi to find jobs and built temporary housing in the suburbs. Although the colonial authority tried to demolish the housing built by Africans under the Vacancy Act and the Public Health Act, it ended up becoming the replica of informal settlements. After independence in 1963, after Africans were released from the movement restrictions imposed on them, Nairobi absorbed people from rural areas seeking job opportunities, thereby expanding the housing needs in Nairobi. Neither public housing provided by the newly independent government nor privately developed housing met the mushrooming housing needs because they were unaffordable for most of the urban population. As a result, most people had no choice but to find shelter in informal settlements, generally in vulnerable areas such as reserved land along rivers and railway lines and low land with high potential for flooding.

Informal settlements have been intermittently bulldozed and the residents forcefully evicted by public authorities and private landowners. With no legal status for informal settlements, the government has been reluctant to address the issue of establishing a legal framework and implementing policy concerning upgrading and formalizing informal settlements. The limited response by the government to deal with the issues of informal settlements is due in part to a lack of political will from unsubstantiated fear that any policy that would foster informal settlements could lead to its further expansion.[Note 2] In addition, any attempt to improve conditions in Nairobi's informal settlements is complicated by potential conflicts between landlords and tenants[Note 3] and by the conflict between different ethnic groups[Note 4] that have often been exacerbated by the manipulations of powerful political interests. Therefore, a new approach to address informal settlements is needed.

Table 1. Basic information about the three villages

Village name	Sisal	Jua Kali[a]	MCC
Area (km²)	0.092	0.0575	0.1013
Population	11,427	6,999	No data
Year of the creation	1985	2000	2000

[a]One area in Vietnam Village.Source: Centre for Urban Research and Innovations, 2013

3. Urban Structure

Nairobi has developed as a center for the service sector industry in Eastern Africa. Maintaining a 5% annual population growth rate since independence, Nairobi had a population of 3.14 million people as of 2009.[16] Nairobi is spatially divided into two parts according to income level: the rich western area, where spacious formal settlements prevail, and the poor eastern area,[17] which is composed of high-density informal settlements, as shown in Figure 1. It should be noted that informal settlements, which comprise 5.2% of the total residential area and house 55%-60% of the total city population, meet the housing needs and form an extremely densely populated residential area, with 50 000 people per sq. km.

4. Methodology

This area of focus is Mukuru Kwa Njenga, which was illegally formed in an industrial zone and houses approximately

Figure 2. Map of Mukuru Kwa Njenga

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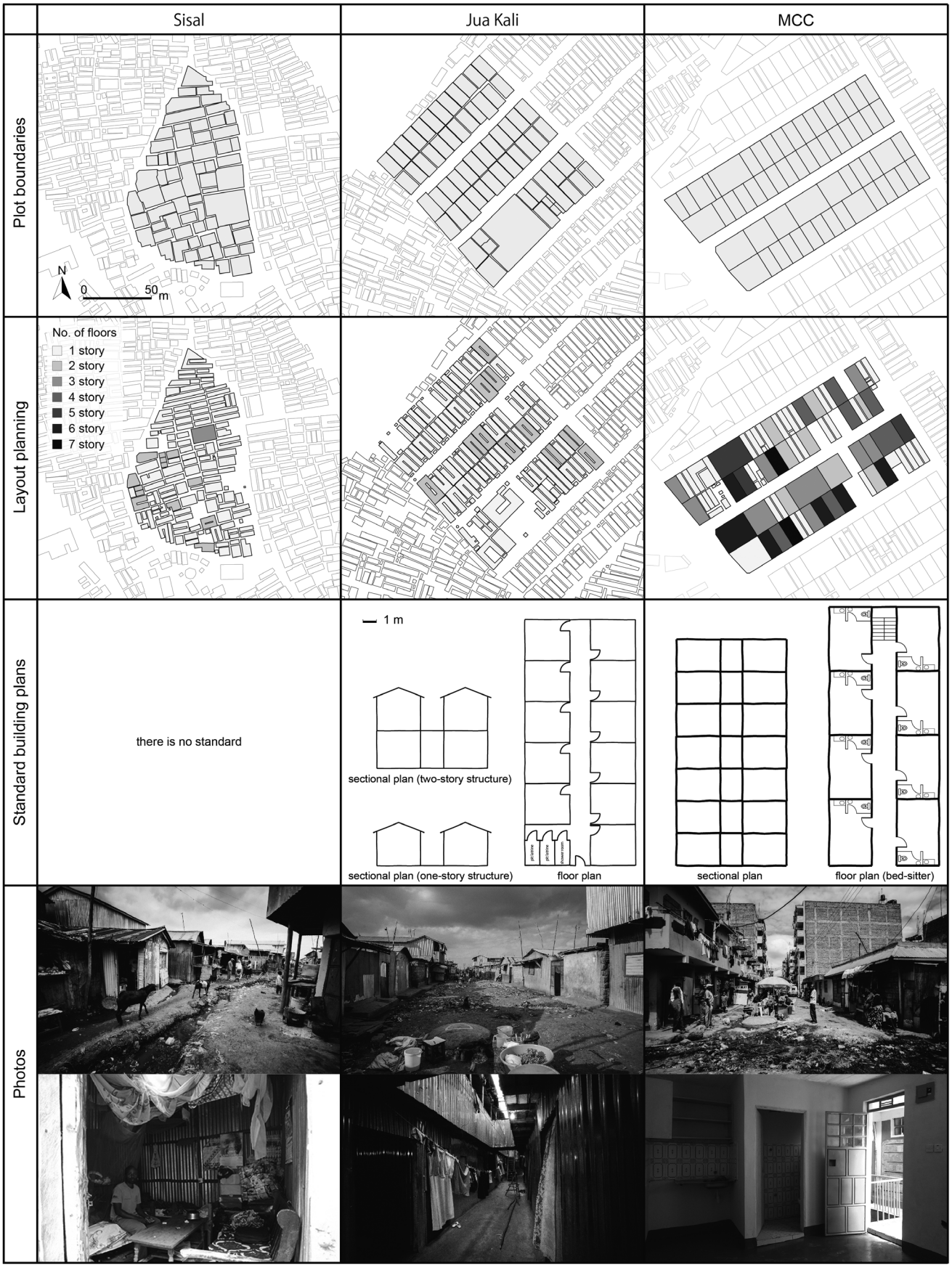

Figure 3. Housing layouts Source: Research findings.

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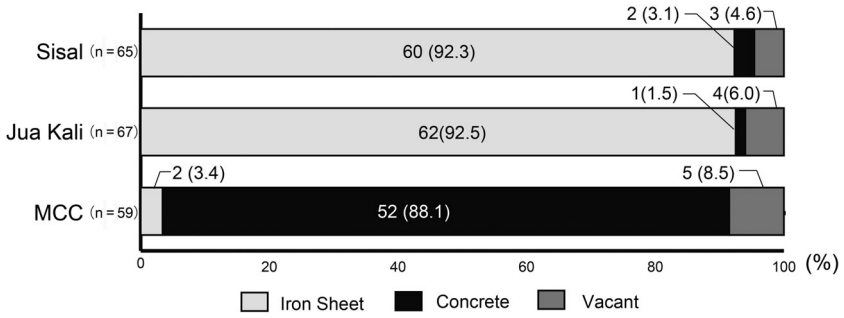

Figure 4. Construction material. Notes: Number shown in parentheses is percent Source: Research findings.

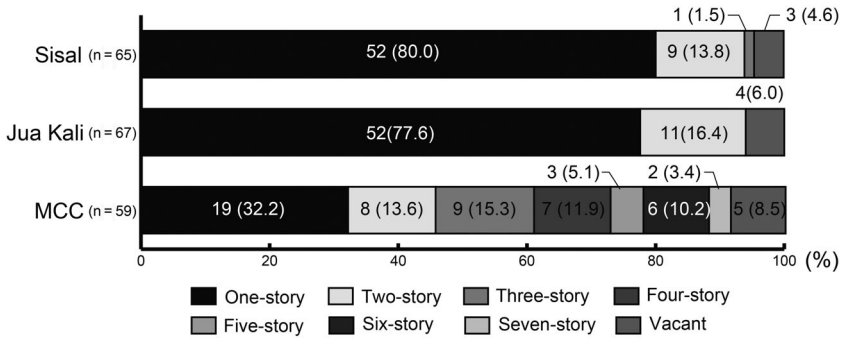

Figure 5. Number of stories. Notes: Number shown in parentheses is percent Source: Research findings.

Table 2. The built environment

	Sisal	Jua Kali	MCC
Number of plots	65	67	59
Gross building coverage ratio (%)[a]	56.8	59.4	64.7
Gross floor area ratio (%)[a]	66.8	71.1	196.7

[a]Calculated within the area surrounded by medial line of streets surrounding the studied area.

150 000 people in a one square kilometer area.[18] The area used to be a large farm owned by white settlers in the colonial era and became an industrial zone after acquisition by the government; the bulk of it was said to be sold to individuals in the 1980s. As most of the buyers owned the land for investment purposes and kept it undeveloped for a long period of time, immigrants from villages, who were seeking job opportunities and housing, came to occupy the land.

As shown in Figure 2, Mukuru Kwa Njenga is divided into eight villages. Sisal, the oldest village in Mukuru Kwa Njenga, started development as early as the 1950s, followed by Milimani, Vietnam, and the northern part of Zone 48, which started development in the 1990s. Then followed Jua Kali in Vietnam village, the southern part of Zone 48, Riara, Wape, Moto, and MCC toward 2000. They have a variety of physical patterns: some villages have temporary structures in a chaotic street pattern, others have high-rise buildings made of concrete blocks arranged in a perfect array. In Nairobi, a settlement with temporary structures made of mud and iron sheet is commonly called a "slum." As Mukuru Kwa Njenga is composed both of "slum" and "formal-looking" settlements, this paper considers both of them together as an "informal settlement," for both are developed outside of any formal development control.

Having different physical patterns, Sisal, Jua Kali, and MCC are chosen as case study areas, to identify the similarities and differences among them. Sisal and MCC coincide within the Sisal village and MCC village, respectively. While Jua Kali, part of Vietnam village, was developed as a block and has a unique development pattern from its surroundings. The basic data about the three areas are shown in Table 1. A series of field studies on the physical patterns, the affordability of housing plots and rooms, and the characteristics of developers was done in 2014. A survey on physical patterns was conducted by means of observation and measurement focusing on site layout, materials, building plans and use, and the built environment. A survey on the affordability of housing plots and rooms, and the characteristics of developers was conducted through interviews with structure owners and residents.

5. Physical Characteristics and Conditions

This section clarifies the physical characteristics and conditions in Mukuru Kwa Njenga, focusing on site layout, building plans, building materials and height, the built environment, and building use.

5.1 Site layout

Figure 3 shows the plot boundaries and layout planning in the areas. Sisal has an unplanned layout with a variety of plot sizes and shapes, while Jua Kali and MCC had planned layouts with 33 by 66 feet uniform strip-shaped plots along a straight

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Photo 1. Medical clinic attached to dwelling Source: Author.

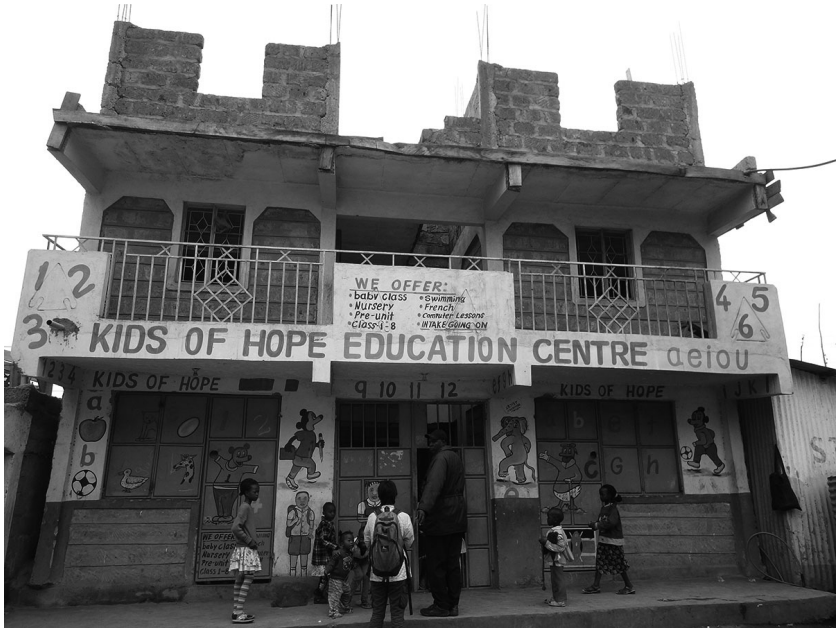

Photo 2. Nursery school attached to dwelling Source: Author.

road with a width of approximately 8 m. As the size and shape of the structures are generally the same as those of the plots, Sisal has structures in a variety of sizes and shapes, while uniform strip-shaped structures are more common in Jua Kali and MCC.

5.2 Building plan

Figure 3 also displays the standard building plans in the areas. As structures in Sisal are varied in size and shape, building plans are varied as well. Though there is no specific standard plan for structures, rectangular-shaped row houses are common. Jua Kali and MCC have uniform plots, and there are standard building plans, as shown in Figure 3. The standard building plan in Jua Kali has a 10 by 12 feet room along and across the aisle, with a shared bathroom and shower room. MCC also has rooms along and across the aisle, and there are some basic types of rooms: a single room, which shares a bathroom and kitchen; a bed sitter, which has a private bathroom and kitchen inside the room; and a one-bed room, which has one bedroom with a private bathroom and kitchen.

5.3 Building material and height

Figure 4 provides the building materials used in the areas. In Sisal and Jua Kali, 92.3% and 92.5% of the structures are made of iron sheet, respectively, while 88.1% of buildings in MCC are made of concrete blocks.

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Figure 5 displays the number of stories of building in the areas. In Sisal and Jua Kali, 80.0% and 77.6% of the structures, respectively, are single-story structure, though there are a few two- and three-story structures. While MCC has a minimum of single-story building and a maximum of seven-story buildings, 45.8% of buildings are low rise (one- to two-story buildings), 27.2% are middle rise (three- to five-story buildings), and 11.5% are high rise (six- to seven-story buildings). As most buildings in MCC are under construction, the area will be lined with high-rise buildings in the near future. In short, low-rise temporary structures made of iron sheet account for the vast majority of structures in Sisal and Jua Kali, while multiple high-rise buildings made of concrete blocks are most common in MCC.

5.4 Built environment

Table 2 shows the built environment in three areas. It shows a 56.8% gross building coverage ratio in Sisal, 59.4% in Jua Kali, and 64.7% in MCC. It also shows a 66.8% gross floor area ratio in Sisal, 71.1% in Jua Kali, and 196.7% in MCC. There is no difference in the gross building coverage ratio among the three areas, but the gross floor area ratio in MCC is higher than those in the other two areas. As MCC has many buildings under construction as well as vacant plots awaiting construction, the gross floor area ratio of MCC is anticipated to stay low. It is expected that almost all the plots in MCC will be filled with seven-story buildings in the near future, which means the gross floor area ratio will be above 500%.

5.5 Building use

In Mukuru Kwa Njenga, there are not only dwellings but also shops, restaurants, schools, clinics, and more, which are necessary for people's daily lives. In terms of business categories, there are retail, services, foods and beverages, entertainment, hotels, manufacturing, banks, medical care, child care and education, religious institutions, and so on. The types of building use can be classified into three categories: dwelling only, dwelling with commerce/business, and commerce/business only. Buildings used for dwelling with commerce/business, generally, have these rooms situated on the ground floor and along the street, and usually set slightly higher rentals for them than the dwelling rooms. Buildings used exclusively for commerce/business have no residents. In Mukuru Kwa Njenga, buildings with a commerce/business use in line with the street constitute a shopping street, which is always full of people shopping and doing business (Photos 1 and 2).

Figure 6 shows the building use in the areas. In Sisal and Jua Kali, buildings for dwelling only account for 70.8% and 61.2%, respectively, and buildings for dwelling with commerce/business account for 21.5% and 29.9%, respectively. It can be said that both areas are specialized residential areas. While in MCC, the ratio of buildings for dwelling only is as low as 25.4%, the ratio of dwelling with commerce/business is 55.9%, and the ratio of commerce/business only is 8.5%; the area is clearly more actively used for commerce/business. This is because MCC is near an arterial road and in a good location for commerce/business; subsequently, the rooms on the ground floor and along the road are made specifically for commerce/business use by default.

6. Affordability of Housing Plots and Rooms

How and by whom are the housing plot and room rentals set in Mukuru Kwa Njenga? Though it is commonly considered that an informal settlement is a living place especially for poor and low-income people who cannot afford the cost of renting or buying formal housing, is it really affordable for those people?

6.1 Affordability of housing plots

Land in Mukuru Kwa Njenga is legally owned by some private individuals and enterprises. But on the ground, the so-called "structure owners" occupy the land, build structures on it, and rent the rooms even without official tenure[Note 5].

Housing plot prices are generally set by the structure owner. There are testimonies from interviews with structure owners that the "land price is set depending on how many rooms can be made on the plot," and the "land price is rising year by year,[Note 6]" thereby indicating the existence of the land market. There are also other testimonies, such as "I bought the land from my friend at a very low price,[Note 7]" and the "village elder[Note 8] gave me a plot for free because my husband was dead and I was very poor finding place to live,[Note 9]" which indicate that the plot price is coordinated in the relationship between sellers and buyers. From Figure 7, which shows the payment amount by structure owners for buying plots in relation to purchase year and plot size, it is clear that the plot price covers a wide range.

In terms of plot affordability, it is far out of reach of the average person in Mukuru Kwa Njenga, whose average daily income is <500 ksh, as a standard-size rectangular plot, that is 33 by 66 feet, costs at least 100 000 ksh (equivalent to about 1000 USD) as of 2014. In other words, the structure owners, who get regular rental incomes from tenants, and own (albeit illegally) the land property, are commonly considered to belong to a relatively high-income class.

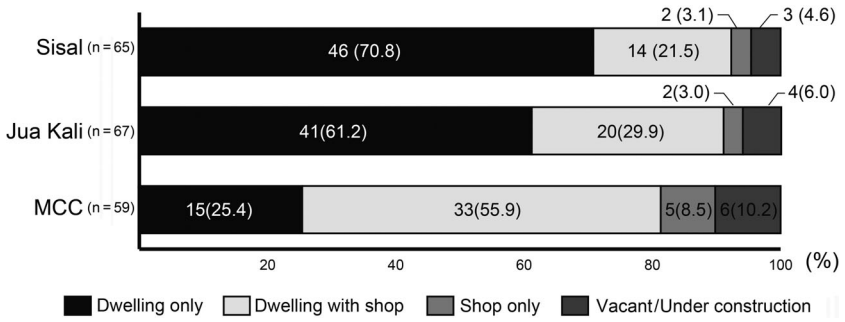

Figure 6. Structure use. Notes: Number shown in parentheses is percent Source: Research findings.

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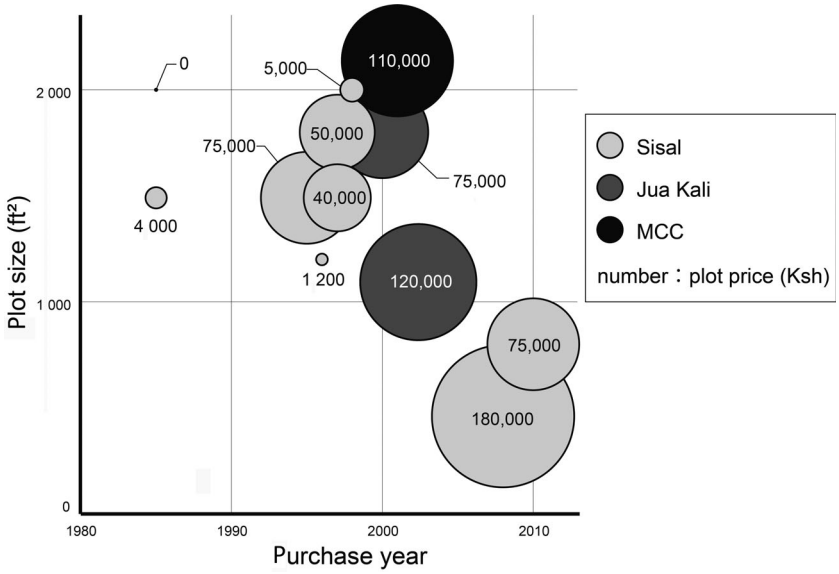

Figure 7. Plot price in relation to purchase year and plot size. Notes: The number of samples: nine plots in Sisal, three plots in Vietnam, one plot in MCC Source: Research findings.

6.2 Affordability of rooms

The rent for a room is also set by the structure owner. Figure 8 displays the room rentals in the three areas. On the whole, rent is dependent on the area: it is generally influenced by the room size, the material of the structure, the available services in the room or on the plot such as water, sanitation, electricity, location (whether situated along the wide street or whether situated in the flooded area), and security of the area.

Sisal is the cheapest rental area in Mukuru Kwa Njenga, where people can rent a room from 500 to 1500 ksh per month; it is located on the lowlands, flooded through almost the whole year, with miserable security problems, and whose typical structures are made of iron sheet, usually without a toilet inside. In Jua Kali, the average room rental ranges from 1500 to 2000 ksh per month, that is slightly higher than that of Sisal; as all the iron-sheet structures in the area are along the wide street and have a shared toilet inside. While almost all of the structures in MCC are made of concrete blocks, situated along the wide street, and with a toilet, water, and electricity inside the structures, MCC is the highest rental area. The rent in MCC varies depending on the room type: a single room costs from 3500 to 5000 ksh per month; a bed sitter costs from 5500 to 7500 ksh per month; and a one bed room costs from 8000 to 12 000 ksh per month.

Considering that the minimum room rental of a formal multi-unit apartment near Mukuru Kwa Njenga costs at least 50 000 ksh per month, the rental market in Mukuru Kwa Njenga supplies a relatively affordable room to the people who cannot afford such formal housing. Furthermore, people can choose a room from a wide range of rented rooms, depending on their income level. It was found from the testimonies of tenants that tenants in Sisal tend to move to Jua Kali, then to MCC, when they become successful in business and earn a better income.

7. Characteristics of Developers

As mentioned before, the people who occupy land and build structures on it without the permission of the legal landowner are commonly known as the structure owners. Figure 9 shows who lives in each plot in the three areas. It shows that in all three areas, there is no plot on which only the structure owner lives, but there are plots with both a structure owner and tenants, and plots with only tenants. The ratio of plots on which a structure owner lives accounts for 26.2% in Sisal, 13.4% in Jua Kali, and 6.8% in MCC, respectively: the low ratio indicates that the structure owners are likely to be absent from the areas in which they build structures, and rather rent them for profit. For most structure owners, the informal settlement is not a place to live but rather an attractive place in which to invest; this tendency is more applicable in MCC than in Sisal.

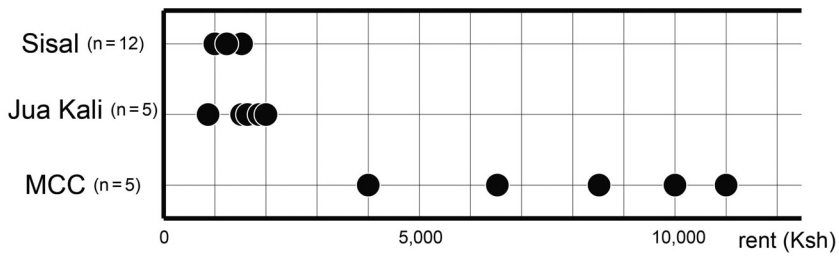

Figure 8. Room rent Source: Research findings.

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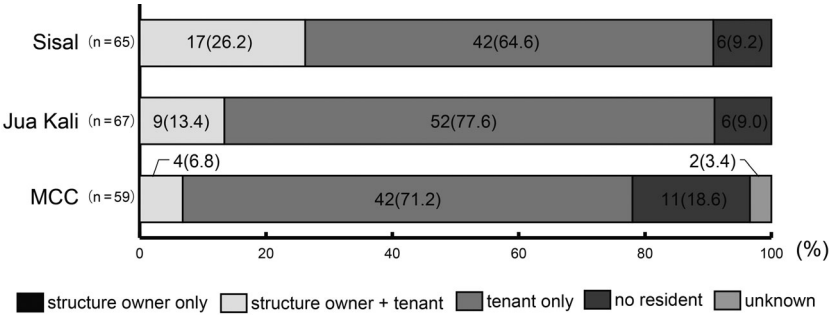

Figure 9. Residents living on plots. Notes: Number shown in parentheses is percent Source: Research findings.

Some structure owners own more than one structure. For example, one structure owner living in Sisal owns two plots in Sisal, one plot in Vietnam, and one in Milimani[Note 10]; another owns three plots in Jua Kali[Note 11]; and yet another owns one plot in MCC and another in a formal settlement in the Nairobi city center.[Note 12] It was found that some structure owners take on completely the characteristics of a developer, buying a plot and building a structure on it as a business opportunity in an informal settlement such as Mukuru Kwa Njenga, while they themselves live in a formal settlement. One structure owner stated, "I'm supplying low-cost housing to meet the needs of poor people, then I gain a small profit from it."

8. Conclusion

The results of our research on the development patterns of informal settlement are as follows:

1 The spatial structure in informal settlements is not uniform. We identified squatter areas with iron-sheet houses and narrow streets, planned areas with high-rise buildings made of concrete blocks arranged in a perfect array, and intermediate conditions.
2 In informal settlements, structures are built and rented to tenants by plot owners who are referred to as structure owners. As a general rule, the plot price and room rental are set by the structure owners. The plot price, which is related to the plot size and the presence or absence of access roads, is unaffordable for most residents. By contrast, room rent, which is related to the size and type of room, the presence or absence of basic services in the room or plot (such as water, sanitation, and electricity), land conditions (such as roadside, low-lying, or flooded areas), and security, is generally fixed in accordance with the area and offers low-income residents a wide range of choices.
3 Few structure owners reside in their own structure, and others are non-residents who build and rent structures for profit. Some own and manage more than one structure. Therefore, informal settlements are not so much a place to live, but rather an attractive investment opportunity for structure owners or developers. This tendency is more applicable to planned areas than in squatter areas in informal settlements.

This research empirically clarifies the diversity of development patterns in informal settlements, using a case study in Nairobi. Further study is needed to identify how these various informal settlements have been formed and managed.

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Disclosures

The authors have no conflict of interest.

Notes

Note 1) It was found out that investment in land and housing, access to formal credit, and municipal revenues have not increased noticeably; there is no significant evidence of poverty levels being reduced; titling does provide increased tenure security; however, many alternative forms of tenure, including those in many informal settlements, also provide high levels of security; land titles do not necessarily protect people from eviction and expropriation of their land; titling does not necessarily improve infrastructure and services provision, while many settlements have obtained improved provision without titles, and so on.

Note 2) Fernandes and Varley showed that the government was concerned any policy that would foster informal development would mean "opening the gates" to more illegality.[13] Although the government runs the risk of being criticized for neglecting the deplorable living standards in informal settlements, it has somewhat succeeded in containing residents' frustrations by providing limited money to improve the environment of informal settlements, such as for road, pavement, and drainage construction, usually in the period preceding an election to attract votes from the inhabitants.

Note 3) Even without official tenure, there are landlords ("structure owners") and tenants, and these two groups have very different priorities within any program to formalize land tenure. Structure owners want to acquire full legal tenure of the land on which their structures are built; tenants want recognition of their right to live there and the possibility of becoming land and house owners.[14] While conflicts between these priorities make it difficult to implement the upgrading program, both parties are concerned about the land tenure formalization in the program. This is because the decision to formalize land tenure could led to land acquisition and robbery by the wealthy and those with political connections in anticipation of rising land prices, structure owners lose their structures, and tenants have difficulty living because of rising rents.

Note 4) The ethnic composition of residents in informal settlements is diverse, but political conflicts complicate relationships between ethnic groups and make it difficult to build consensus on environmental improvements, [15] as seen in the 2007–2008 post-election crisis caused by presidential candidates from different ethnicities.

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Note 5) The land of Mukuru Kwa Njenga used to be a large farm owned by white settlers in the colonial era and became an industrial zone after acquisition by the government; the bulk of it was said to be sold to individuals in the 1980s. As most of the buyers have owned the land for investment purposes and kept it undeveloped for a long period of time, immigrants from villages, who were seeking job opportunities and housing, came to occupy the land.

Note 6) By interview with the village elder in Sisal (May 14th, 2014).

Note 7) By interview with the structure owner in Sisal (March 25th, 2014).

Note 8) A village elder called "chief," appointed by a government official, leads a village committee to maintain the order in the village.

Note 9) By interview with the structure owner in Sisal (June 10th, 2014).

Note 10) By interview with the structure owner in Sisal (June 10th, 2014).

Note 11) By interview with the structure owner in Jua Kali (July 5th, 2014).

Note 12) By interview with the structure owner in MCC (July 3rd, 2014).

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